



Operating Instructions

Diesel engine

12V2000S52 / R1238K40-1820

12V2000S62 / R1238K40-1822

16V2000S52 / R1638K40-1815

16V2000S52R / R1638K40-1814

Application group 4B

MS150042/05E

Engine model	Application group
12V2000S52	4B Continuous operation, variable
12V2000S62	4B Continuous operation, variable
16V2000S52	4B Continuous operation, variable
16V2000S52R	4B Continuous operation, variable

Table 1: Applicability

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California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.



WARNING: Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1	Preface		6.10 Plant - Cleaning	41
1.1	Preface	7	7 Maintenance	
2	Safety		7.1 Maintenance task reference table [QL1]	43
2.1	Important provisions for all products	8	8 Troubleshooting	
2.2	Intended use of all products	9	8.1 Troubleshooting	44
2.3	Personnel and organizational requirements	10	8.2 Fault messages from DDEC engine governor III/IV	47
2.4	Safety regulations for commissioning and operation	11	9 Task Description	
2.5	Safety regulations for assembly, maintenance, and repair work	13	9.1 Engine	66
2.6	Fire and environmental protection, indirect materials	17	9.1.1 Engine - Barring manually	66
2.7	Standards for warning messages in the text and highlighted information	20	9.1.2 Engine - Turning on starting system	67
3	Transport		9.2 Cylinder Liner	68
3.1	Transport	21	9.2.1 Cylinder liner - Endoscopic examination	68
4	General Information		9.2.2 Cylinder liner - Instructions and comments on endoscopic and visual examination	70
4.1	Engine side and cylinder designations	22	9.3 Crankcase Breather	72
4.2	Engine layout	23	9.3.1 Crankcase breather - Oil separator replacement, diaphragm check and replacement	72
4.3	Sensors - Overview	25	9.3.2 Crankcase breather - Wire mesh cleaning	74
4.4	Data acquisition and data transmission	26	9.4 Valve Drive	76
5	Technical Data		9.4.1 Valve clearance - Check and adjustment	76
5.1	12V 2000 S52, 12V 2000 S62, 16V 2000 S52, 16V 2000 S52R engine data	27	9.4.2 Cylinder head cover - Removal and installation	78
5.2	Firing order	30	9.5 Injection Pump / HP Pump	79
5.3	Engine - Main dimensions	31	9.5.1 Injection pump - Replacement	79
6	Operation		9.5.2 Injection pump - Removal and installation	80
6.1	Preparations for putting into operation after extended out-of-service periods (>3 months)	32	9.6 Injection Valve / Injector	84
6.2	Preparations for putting into operation after scheduled out-of-service-period	33	9.6.1 Injector - Replacement	84
6.3	Engine - Starting	34	9.6.2 Injector - Removal and installation	85
6.4	Re-starting the engine following an automatic safety shutdown	35	9.7 Fuel System	90
6.5	Safety system - Override	36	9.7.1 HP fuel line and pressure pipe neck - Replacement	90
6.6	Operational checks	37	9.7.2 Fuel - Draining	92
6.7	Engine shutdown	38	9.7.3 Fuel system - Venting	93
6.8	Emergency engine stop	39	9.8 Fuel Filter	98
6.9	After stopping the engine - Putting the engine out of operation	40	9.8.1 Fuel filter - Replacement	98
			9.9 Charge-Air Cooling	99
			9.9.1 Intercooler - Checking condensate drain for coolant leakage and obstruction	99
			9.10 Air Filter	100
			9.10.1 Air filter - Replacement	100

9.11 Starting Equipment	101	9.15.4 Charge-air coolant - Filling	118
9.11.1 Starter - Condition check	101	9.15.5 Charge-air coolant pump - Relief bore check	120
9.12 Lube Oil System, Lube Oil Circuit	102	9.16 Wiring (General) for Engine/Gearbox/Unit	121
9.12.1 Engine oil level - Check	102	9.16.1 Engine cabling - Check	121
9.12.2 Engine oil - Change	103	9.17 Accessories for (Electronic) Engine Governor / Control System	122
9.13 Oil Filtration / Cooling	105	9.17.1 Engine governor and connectors - Cleaning	122
9.13.1 Engine oil filter - Replacement	105	9.17.2 Engine governor plug connections - Check	123
9.14 Coolant Circuit, General, High-Temperature Circuit	106	9.17.3 DDEC engine governor - Removal and installation	124
9.14.1 Engine coolant - Level check	106	10 Appendix A	
9.14.2 Engine coolant - Change	108	10.1 Manufacturer's documentation	125
9.14.3 Engine coolant - Draining	109	10.2 Contact person/Service partner	126
9.14.4 Engine coolant - Filling	110	10.3 Abbreviations	127
9.14.5 Engine coolant pump - Relief bore check	112	11 Appendix B	
9.14.6 Engine coolant - Sample extraction and analysis	113	11.1 Standard Tools and Special Tools	130
9.15 Low-Temperature Circuit	114	11.2 Index	136
9.15.1 Charge-air coolant level - Check	114		
9.15.2 Charge-air coolant - Change	116		
9.15.3 Charge-air coolant - Draining	117		

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:<http://www.mtu-solutions.com>

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.